

adesso.it

From «Hello, world!» to distributed systems

A practical overview of the key points to master if you want to be a good dev!

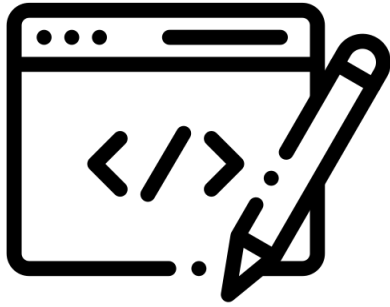
Jacopo Di Stanislao
Software Architect @ adesso.it



Agenda

1. **A software developer journey**
2. **Well coded software**
3. **Demo**
4. **Well designed software**
5. **Demo**
6. **Q&A**

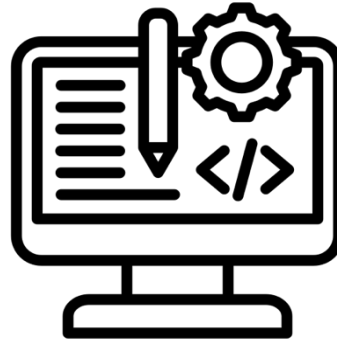
○ A software developer journey



Junior
Developer



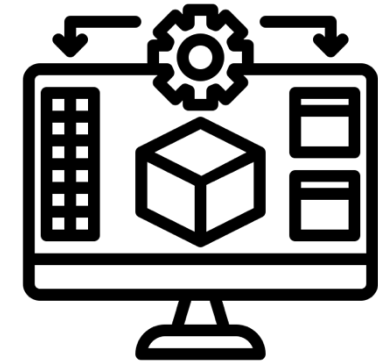
Code design



Senior
Developer



Software design



Software
Architect



System design

○ Well coded software

// Clear

// Evolvable

// Correct

// Simple

Well coded software

// Clear

- Classes
- Methods
- Variables
- Types
- Return types
- Error handling
- if/else/switch
- $O(n)$

// Evolvable

// Correct

// Simple

Well coded software

// Clear

- Classes
- Methods
- Variables
- Types
- Return types
- Error handling
- if/else/switch
- $O(n)$

// Evolvable

- Encapsulation
- Cohesion vs Coupling
- SOLID
- Design Patterns

// Correct

// Simple

Well coded software

// Clear

- Classes
- Methods
- Variables
- Types
- Return types
- Error handling
- if/else/switch
- $O(n)$

// Evolvable

- Encapsulation
- Cohesion vs Coupling
- SOLID
- Design Patterns

// Correct

- Unit testing
- Integration testing
- Code reviews
- TDD

// Simple

Well coded software

// Clear

- Classes
- Methods
- Variables
- Types
- Return types
- Error handling
- if/else/switch
- $O(n)$

// Evolvable

- Encapsulation
- Cohesion vs Coupling
- SOLID
- Design Patterns

// Correct

- Unit testing
- Integration testing
- Code reviews
- TDD

// Simple

- DRY
- KISS
- Fits in your head
- Good enough

Well coded software

“Talk is cheap, show me the code!”
(*Linus Torvalds*)

Well designed software

// Deployment
Model

// Sync vs.
Async

// Distributed
Systems

// Observability

Well designed software

// Deployment Model

- Monolith
- Microservices

// Sync vs. Async

// Distributed Systems

// Observability

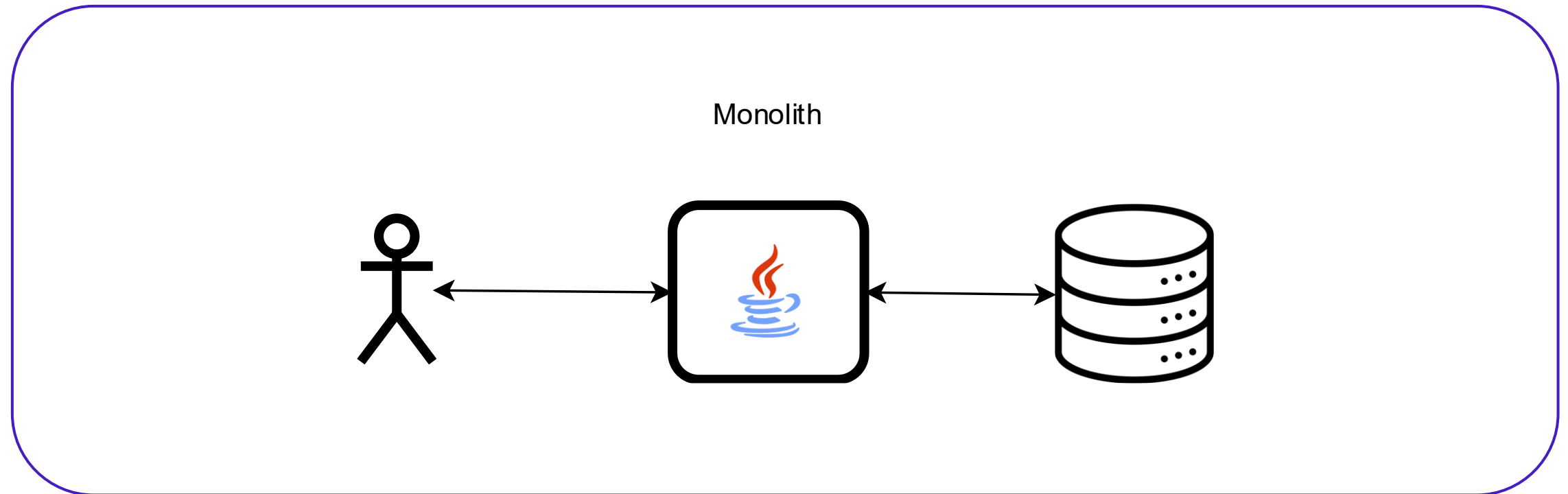
Well designed software

// **Deployment Model**

// Sync vs. Async

// Distributed Systems

// Observability



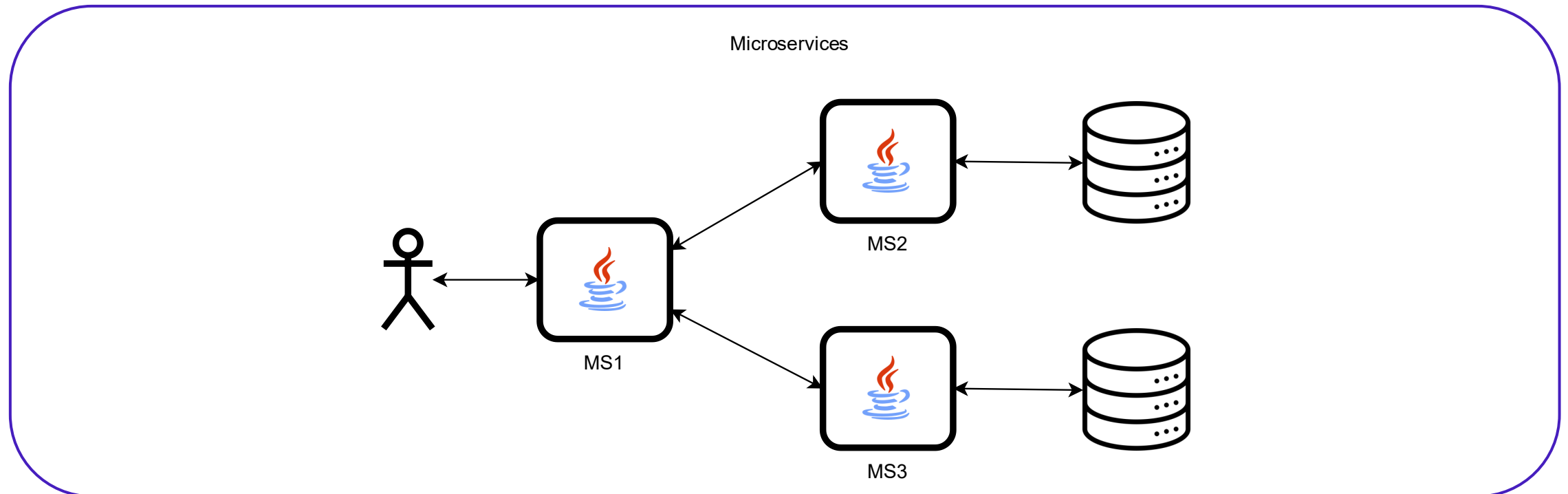
Well designed software

// **Deployment Model**

// Sync vs. Async

// Distributed Systems

// Observability



Well designed software

// Deployment Model

- Monolith
- Microservices

// Sync vs. Async

- API
- Messages
- Idempotency

// Distributed Systems

// Observability

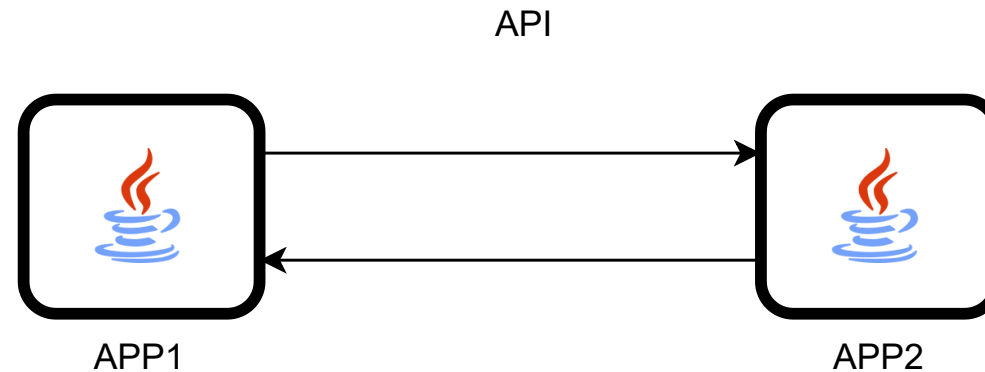
Well designed software

// Deployment Model

// Sync vs. Async

// Distributed Systems

// Observability



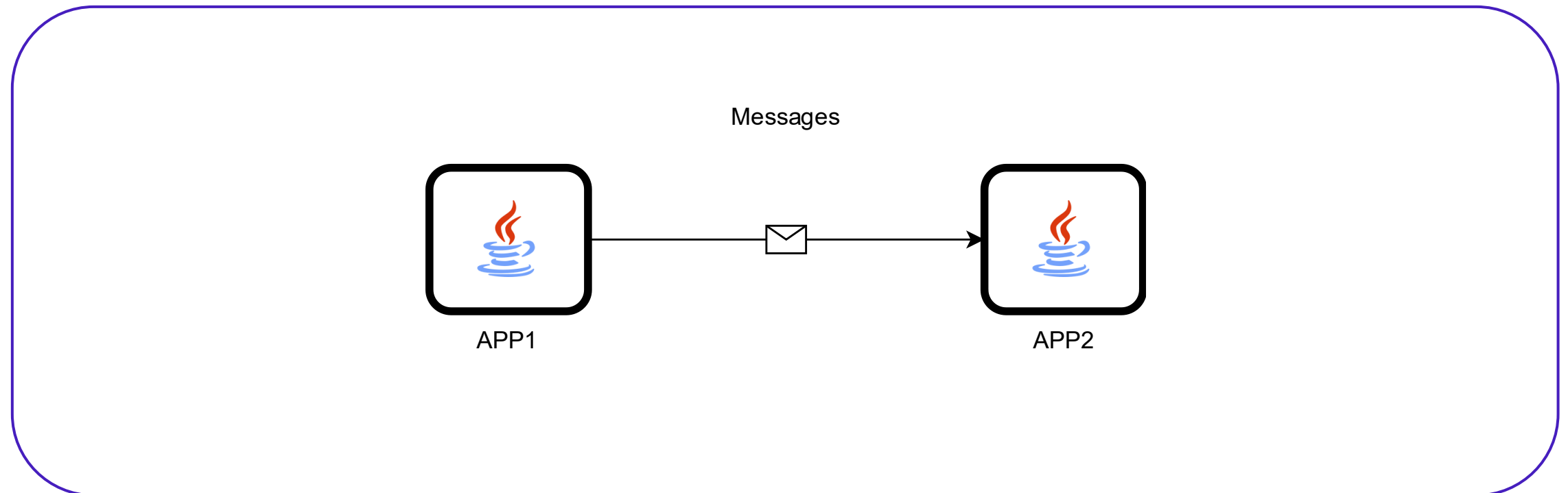
Well designed software

// Deployment Model

// Sync vs. Async

// Distributed Systems

// Observability



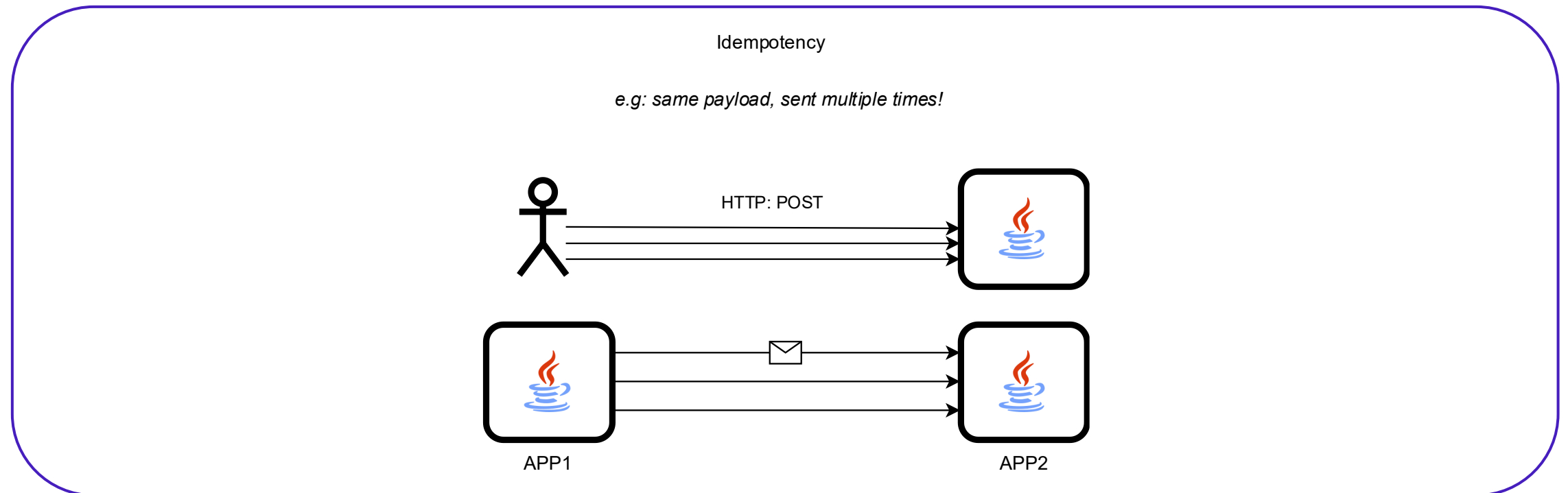
Well designed software

// Deployment Model

// Sync vs. Async

// Distributed Systems

// Observability



Well designed software

// Deployment Model

- Monolith
- Microservices

// Sync vs. Async

- API
- Messages
- Idempotency

// Distributed Systems

- Transactions
- Errors
- Eventual consistency

// Observability

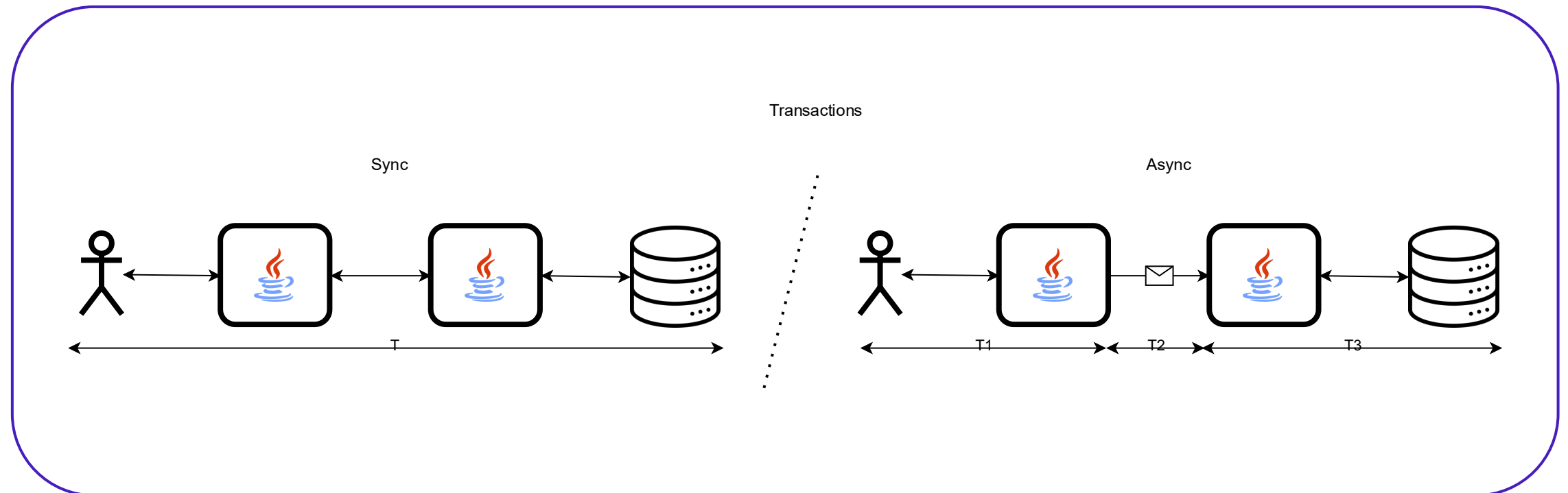
Well designed software

// Deployment Model

// Sync vs. Async

// Distributed Systems

// Observability



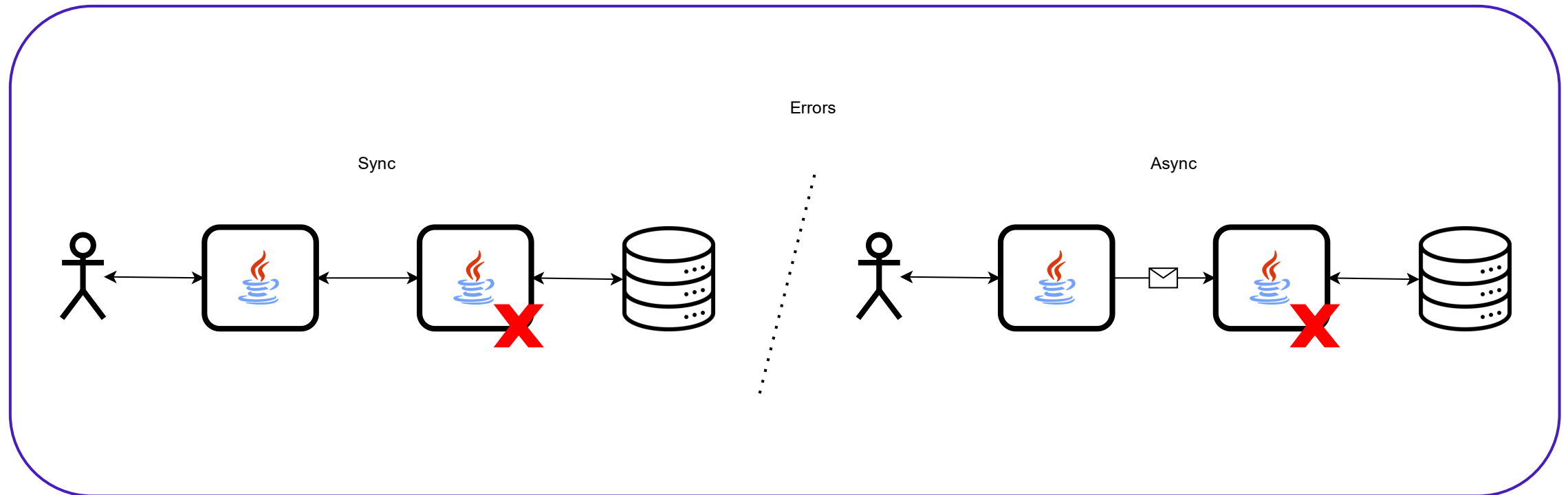
Well designed software

// Deployment Model

// Sync vs. Async

// Distributed Systems

// Observability



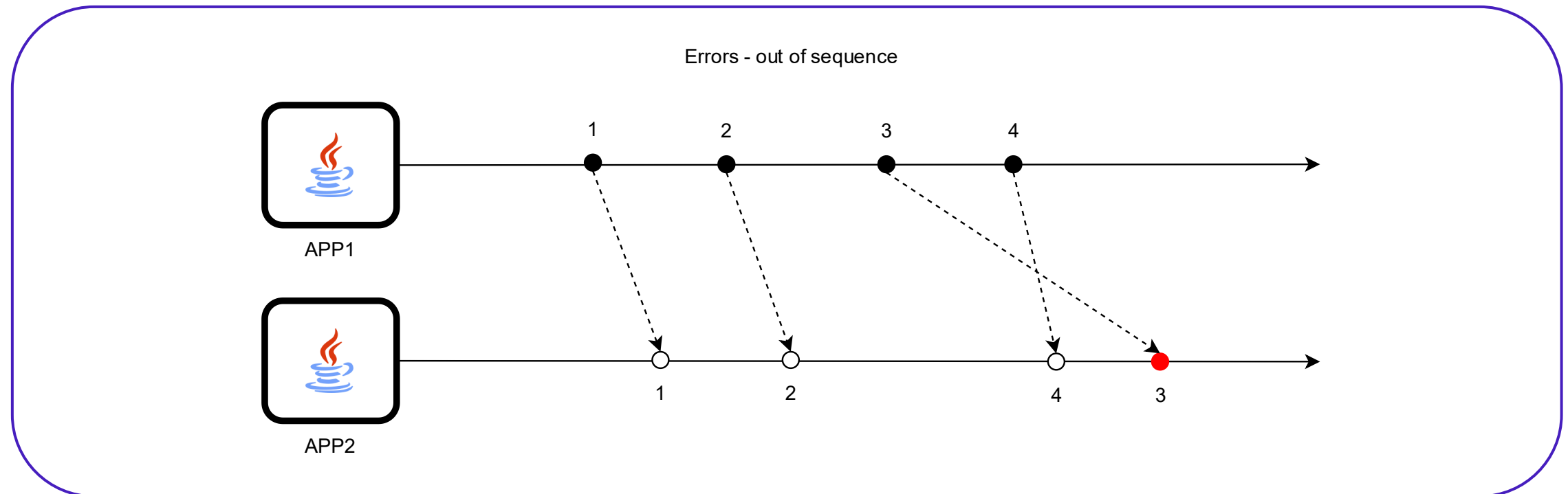
Well designed software

// Deployment Model

// Sync vs. Async

// **Distributed Systems**

// Observability



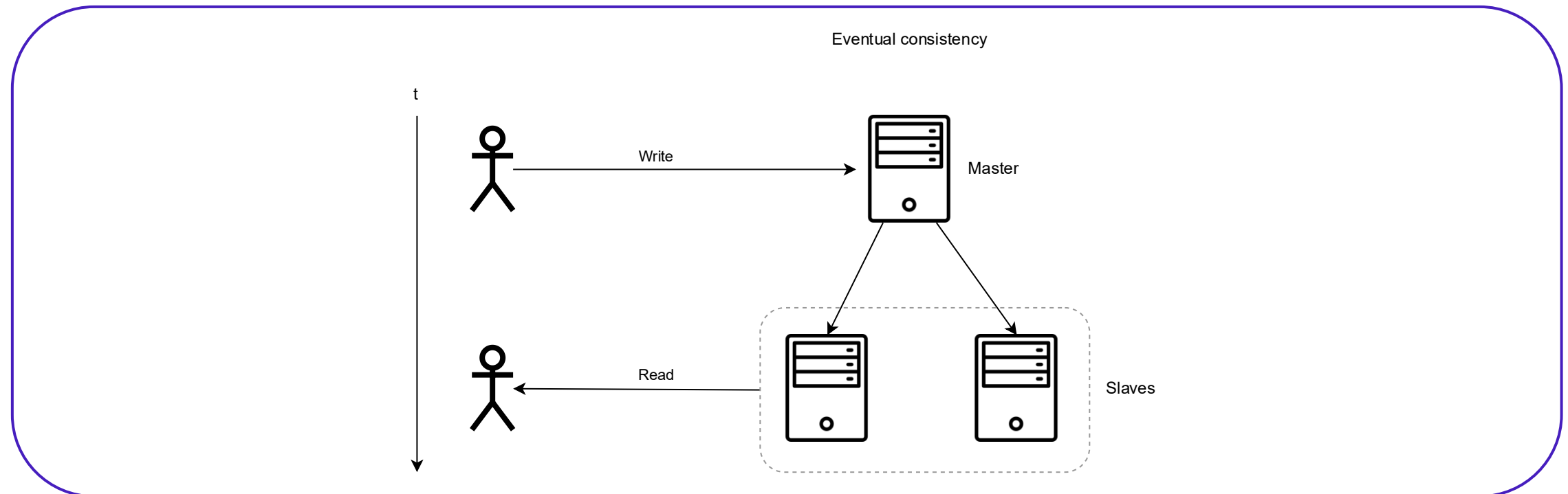
Well designed software

// Deployment Model

// Sync vs. Async

// **Distributed Systems**

// Observability



Well designed software

// Deployment Model

- Monolith
- Microservices

// Sync vs. Async

- API
- Messages
- Idempotency

// Distributed Systems

- Errors
- Transactions
- Eventual consistency

// Observability

- Logs
- Metrics
- Tracing

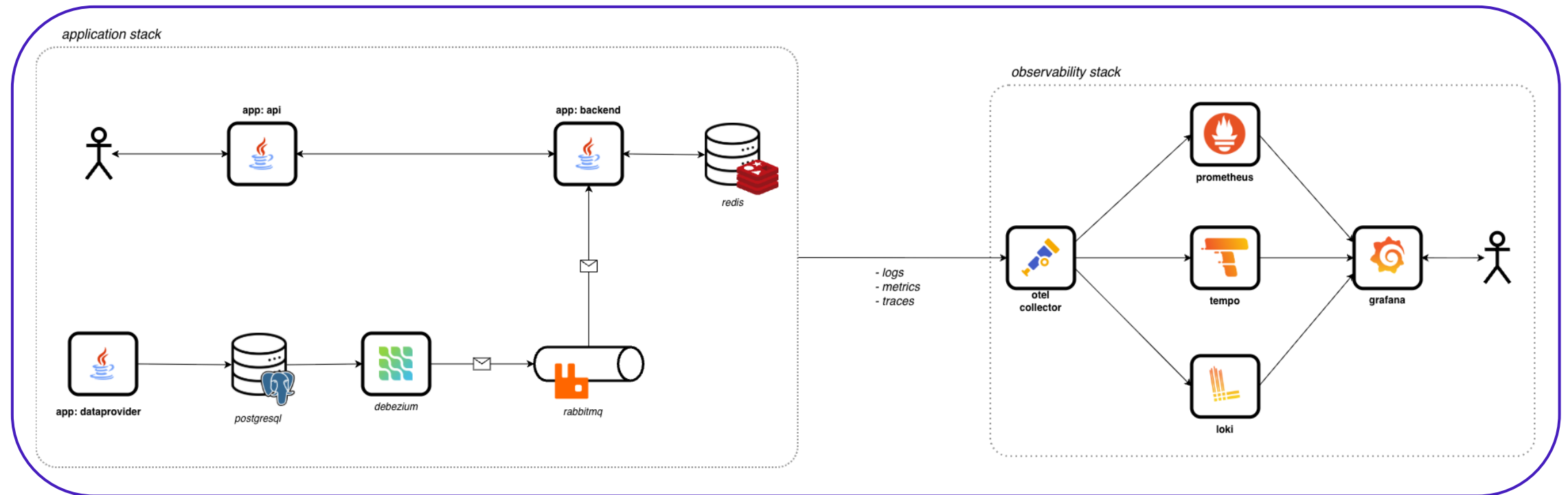
Well designed software

// Deployment Model

// Sync vs. Async

// Distributed Systems

// **Observability**



Well designed software

“Demo!”

Q&A



Demo 1 repo

<https://tinyurl.com/5742mxar>

Grazie mille.

 www.linkedin.com/company/adesso-it

 www.adesso.it

 info@adesso.it

MILANO

Viale Edoardo Jenner, 51 20159
Milano, Italia

+39 02 697731

LECCE

Via Merine, 36 73100
Lecce, Italia

+39 0832 1778061

adesso.it